

FREMONT, MI, USA

WMO: 720415

Lat: 43.433N

Lon: 86.000W

Elev: 253

StdP: 98.32

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00140

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.7	-22.4	0.5	-14.1	-19.7	0.7	-11.2	11.0	-1.7	9.4	-4.1	1.9	270	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	31.1	21.8	29.6	21.2	27.9	20.1	23.8	28.1	22.9	27.2	22.1	25.7	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.9	25.5	22.0	17.2	24.7	21.1	16.2	24.0	72.5	28.2	69.3	26.9	66.1	25.2	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.4	7.5	-22.0	33.5	3.8	1.7	-24.7	34.7	-26.9	35.7	-29.0	36.7	-31.7	38.0		
(5)			-22.2	25.5	3.8	1.0	-25.0	26.2	-27.2	26.8	-29.3	27.4	-32.0	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-5.1	-4.1	1.0	7.3	14.4	19.2	21.4	20.4	16.8	10.0	3.6	-1.4	(6)
(7)		DBStd	10.44	5.57	5.68	5.85	4.86	4.87	3.35	3.27	2.92	4.06	4.56	4.80	4.76	(7)
(8)		HDD10.0	1886	468	395	287	109	13	0	0	0	2	59	199	353	(8)
(9)		HDD18.3	3811	727	628	538	332	141	29	10	14	77	262	442	612	(9)
(10)		CDD10.0	1409	0	0	7	28	151	276	354	321	206	58	8	0	(10)
(11)		CDD18.3	291	0	0	0	1	19	54	106	77	31	2	0	0	(11)
(12)		CDH23.3	2491	0	0	4	11	204	458	979	588	226	19	0	0	(12)
(13)		CDH26.7	642	0	0	1	0	51	109	304	126	51	1	0	0	(13)
(14)	Wind	WSAvg	3.0	3.4	3.5	3.4	3.8	3.1	2.8	2.3	2.1	2.4	2.9	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	889	59	48	61	81	101	92	80	78	77	87	83	59	(15)
(16)		PrecMax	1101	99	99	130	132	251	152	185	141	181	196	183	137	(16)
(17)		PrecMin	741	21	20	12	22	19	38	32	22	5	6	25	29	(17)
(18)		PrecStd	111	28	21	33	31	56	36	35	36	40	52	48	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	13.3	23.9	25.0	31.0	31.9	33.5	31.3	31.2	26.2	19.7	13.9	(19)
(20)			MCWB	8.2	9.7	17.5	14.5	20.8	23.6	23.5	22.0	20.7	17.6	13.0	12.2	(20)
(21)		2%	DB	6.2	9.2	16.3	22.0	27.8	29.6	31.4	29.7	27.9	22.6	16.2	9.1	(21)
(22)			MCWB	4.4	6.7	12.4	12.3	19.1	20.2	22.1	21.6	20.7	16.6	11.5	7.7	(22)
(23)		5%	DB	3.5	5.4	12.8	18.5	25.5	27.7	29.9	27.9	26.0	20.1	12.9	6.9	(23)
(24)			MCWB	2.4	3.1	8.7	10.8	16.7	19.3	21.7	20.6	18.8	15.8	9.5	5.7	(24)
(25)		10%	DB	2.1	2.9	9.7	15.8	23.1	26.1	28.0	26.8	24.0	17.7	11.0	4.7	(25)
(26)			MCWB	1.1	1.2	6.0	9.5	16.1	18.5	20.4	19.9	18.0	14.3	8.1	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.9	17.5	17.4	22.4	24.7	25.4	24.1	23.7	20.4	15.0	13.2	(27)
(28)			MCDWB	9.9	11.3	23.4	21.6	28.4	29.6	30.2	27.6	27.6	22.5	17.5	13.6	(28)
(29)		2%	WB	4.6	6.8	13.9	14.3	20.6	22.5	24.2	23.3	22.3	18.6	12.2	8.0	(29)
(30)			MCDWB	5.6	9.7	15.3	18.1	25.3	26.9	28.7	27.0	25.6	20.3	15.0	8.8	(30)
(31)		5%	WB	2.6	3.5	9.1	12.6	19.1	21.3	23.2	22.5	21.2	16.8	10.1	5.8	(31)
(32)			MCDWB	3.6	5.4	11.9	16.8	23.5	25.0	27.7	25.9	24.0	19.3	13.0	6.6	(32)
(33)		10%	WB	1.2	1.6	6.2	10.3	17.6	20.3	22.3	21.6	20.0	14.7	8.1	3.3	(33)
(34)			MCDWB	1.9	2.9	9.2	14.6	21.3	23.9	26.2	24.5	22.9	17.4	10.2	4.5	(34)
(35)	Mean Daily Temperature Range	MDBR	6.6	7.6	9.1	10.7	11.7	11.1	11.6	11.0	10.9	9.4	7.2	5.5	(35)	
(36)		5% DB	MCDBR	7.7	9.6	12.8	15.1	14.1	13.0	12.9	12.3	12.5	12.8	10.8	8.7	(36)
(37)			MCWBR	6.7	7.4	8.1	7.7	6.5	5.7	5.3	5.3	5.8	7.3	7.1	7.7	(37)
(38)		5% WB	MCDWB	7.5	9.1	11.3	12.3	11.7	10.3	10.9	9.9	9.7	10.5	9.7	8.5	(38)
(39)			MCWBR	6.8	7.8	8.2	8.4	6.4	5.8	5.4	5.1	5.5	7.9	7.5	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.319	0.338	0.366	0.398	0.414	0.410	0.406	0.381	0.346	0.313	0.288	(40)	
(41)		taud	2.369	2.289	2.363	2.318	2.286	2.288	2.298	2.313	2.361	2.454	2.487	2.437	(41)	
(42)		Ebn at Noon	837	870	901	898	874	859	857	849	844	833	812	812	(42)	
(43)		Edh at Noon	79	101	107	122	130	130	127	121	107	86	71	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.22	2.07	3.57	4.69	5.54	6.18	6.28	5.33	4.25	2.61	1.52	0.98	(44)	
(45)		RadStd	0.13	0.24	0.38	0.44	0.52	0.39	0.32	0.33	0.37	0.32	0.21	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page